

Data availability & access, quality tests, stationXML, AdA webpage

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INSTITUTE OF GEOPHYSICS
OF THE CZECH ACADEMY OF SCIENCES



station names

temporary deployments in new sites

stations names of new stations should read country + number + version (A/B/..), example BHxyA for Bosnia and Herzegovina, all these stations names will have 5 characters.

exception: Greece GRxy ... without A at the end, only 4 characters

temporary upgrades of existing stations

station names of the “upgraded” SP/SM permanent stations should keep the permanent station name

- sometimes yes (Albania, Bulgaria, Romania,...), sometimes no (Bosnia and Herzegovina)

station names of the known sites, where there was / is station before:

- former AlpArray = keep the name of AlpArray (unequipped for the moment)

- former PACASE = keep the name of PACASE (running, some built for PACASE following already country codes [SK18A], some taken from AlpArray-PACASE, example A001A)

exceptions: UniTwente stations deployed by INGV in Italy, “almost” AlpArray sites, named ITxyA; the places are in fact different, sometimes however very close and hence for many methods to be used as the same station

there are also **stations deployed before** this agreement, and later merged to AdA, like those by Croatian Seismological Survey in Croatia, having different names, let's keep these already given

location code (default): 00

minor site changes (<10m) => 01, 02, etc.

major site changes (>10m) => last character of station name (default A) changed to B, C, etc.

after minor and major site changes the metadata have to be updated

network codes

- temporary (open or emargoes) vs. permanent (open)

temporary

- idea: one code for temporary stations connected to a given EIDA node (one node – one code)
- in practice: one code goes to one node, more codes go to one node (INGV)

temporary deployments in new sites:

- new station will have new temporary FDSN network code

temporary upgrades of existing stations already in EIDA

- option 1: The same FDSN code as the permanent station, difference only by channel
 - stations from this network MUST be distributed from the same EIDA node as before (e.g. AC in Albania from EIDA@INGV)
 - all channels from this station are fully open without restrictions
- option 2: a new FDSN code as for other temporary stations in given region/EIDA node

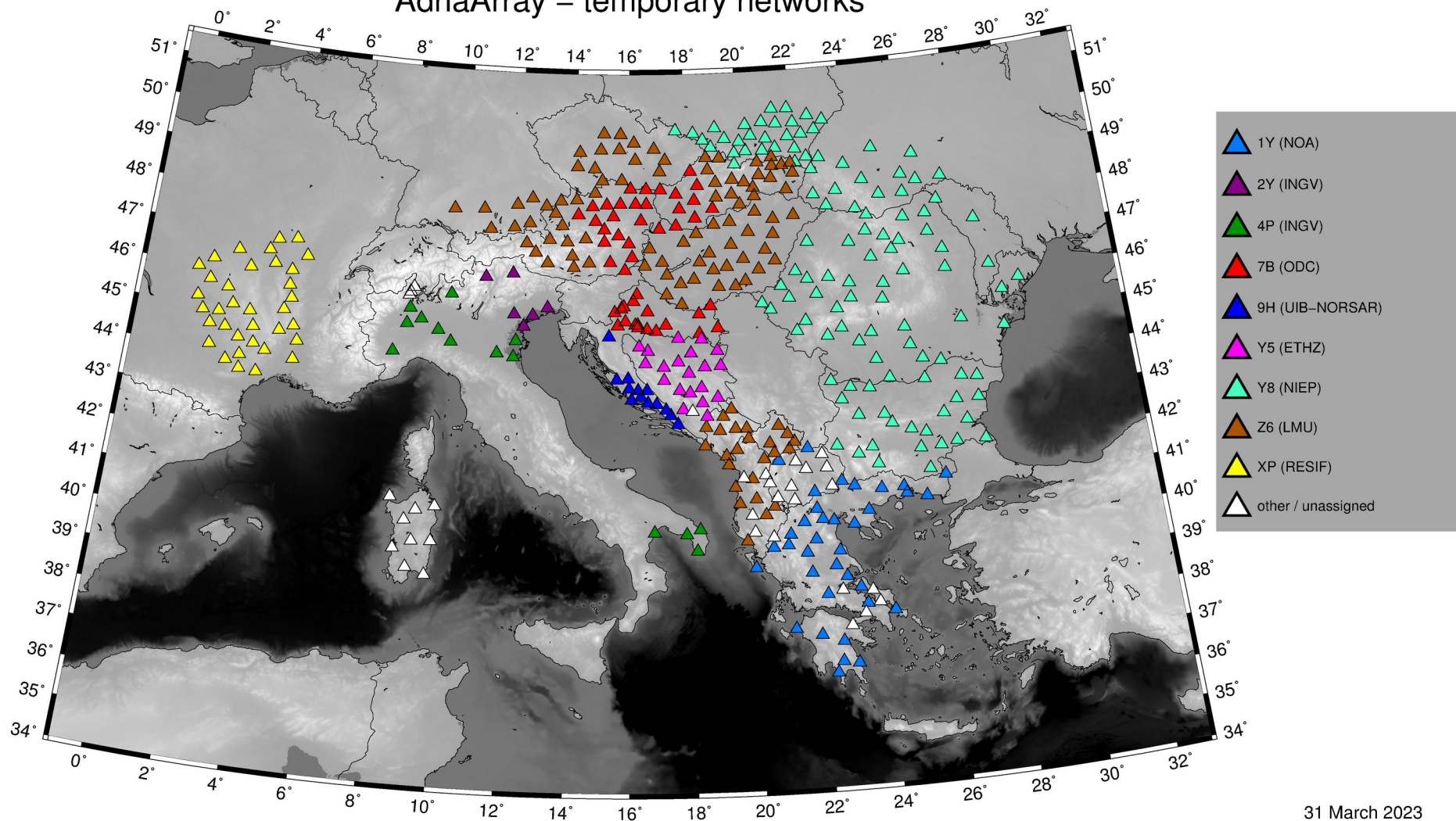
temporary upgrades of existing stations NOT in EIDA yet

- option 1: the same code as other permanent station from the same FDSN network
 - stations from this network MUST be distributed from the same EIDA node as possibly other stations already distributed from the same network (e.g. BS in Bulgaria from EIDA@NIEP)
- option 2: a new code as for other temporary stations in given region/EIDA node

Network codes for the new installed stations were set up independently for each EIDA node. After checking the possible options, we gave up on having a set of similar network codes, as these are not available following the rules for assigning the codes to the temporary experiments.

	network		open /	country of
EIDA node	code	mobile pool	/ embargo	deployment
NOA	1Y	DSEBRA - Uni Bochum	embargo	Greece
				North Macedonia
INGV	2Y	OGS	embargo	Italy
INGV	4P	INGV Bologna	open	Italy
		Uni Twente	open	Italy
		Barcelona	open	Italy
ODC	7B	UniWien	open	Austria
				Slovakia
		Croatian Seism. Survey	open	Croatia
UIB-NORSAR	9H	Uni Zagreb+Norwegian Pool	embargo	Croatia
				BiH
RESIF	XP	Resif-Sismob	open	France
ETHZ	Y5	ETH	open	BiH
NIEP	Y8	MOBNET IG CAS CZ	embargo	Romania
				Bulgaria
		Uni Aarhus	embargo	Romania
				Bulgaria
		Uni Oulu	embargo	Romania
		Uni Helsinki		Romania
		Polish AdA Group	embargo	Poland
				Ukraine
		NIEP		Moldova
		IRSM CAS CZ/Uni Helsinki	embargo	Romania
LMU	Z6	MOBNET IG CAS CZ	embargo	Slovakia
				Czech Republic
		DSEBRA - Kiel	embargo	Germany
				Austria
				Hungary
		EPSS Hungary	embargo	Hungary
		DSEBRA - LMU	embargo	Albania
				Montenegro
				Kosovo
		Montenegro pool	embargo	Montenegro
		Kosovo pool	embargo	Kosovo

AdriaArray – temporary networks



1Y	AdriaArray Temporary Network: Greece, North Macedonia
2Y	AdriaArray Temporary Network: Italy - northwest
4P	AdriaArray Temporary Network: Italy - north, south
7B	AdriaArray Temporary Network: Austria, Croatia, Slovakia
9H	AdriaArray Temporary Network: Croatia
XP	AdriaArray Temporary Network: France
Y5	AdriaArray Temporary Network: Bosnia and Herzegovina
Y8	AdriaArray Temporary Network: Bulgaria, Moldova, Poland, Romania, Ukraine
Z6	AdriaArray Temporary Network: Albania, Austria, Czech Rep., Germany, Hungary, Kosovo, Montenegro, Slovakia

Network Code	Network Name	Operated By	Deployment	DOI
1Y (2022-2026)	Greece/North Macedonia Contribution to AdriaArray Temporary Network	 National Observatory of Athens, Institute of Geodynamics, Athens	—	—
2Y (2022-2026)	OGS Contribution to AdriaArray Temporary Network	 National Institute of Oceanography and Applied Geophysics (OGS)	—	—
4P (2022-2026)	INGV/UniTwente/Geo3Bcn-CSIC Contribution to AdriaArray Temporary Network	 Istituto Nazionale di Geofisica e Vulcanologia (INGV)	—	—
7B (2022-2026)	AdriaArray Temporary Network: Austria, Croatia, Slovakia	 Croatian Seismological Survey (CSS)  ORFEUS (KNMI) Data Center, Royal Netherlands Meteorological Institute  UniWien (Department of Meteorology and Geophysics)	—	—
9H (2022-2026)	CRONOS – Croatia/Norway Contribution to AdriaArray Temporary Network	Norwegian Pool of Advanced Broad-Band Seismic Instruments  University of Bergen (UiB Norway)  University of Zagreb	—	—
XP (2023-2027)	AdriaArray Temporary Network: France – MACIV backbone (Multiscale seismic imaging of Massif Central)	 ISTERre (ISTERRE)  Research Institute in Astrophysics and Planetology (IRAP France)	—	—
Y5 (2022-2026)	Swiss Contribution to AdriaArray Temporary Network	 Eidgenössisch Technische Hochschule (ETH Zurich) – Swiss Seismological Service	—	—
Y8 (2022-2026)	NIEP Contribution to AdriaArray Temporary Network	 National Institute for Earth Physics (NIEP Romania)	—	—
Z6 (2022-2026)	LMU Contribution to AdriaArray Temporary Network	 Ludwig Maximilians Universität München (LMU-Germany)	—	—

	institution
Operated By	1Y National Observatory of Athens, Institute of Geodynamics, Athens
	Aristotle University of Thessaloniki
	University of Patras
	University of Athens
	Saints Cyril and Methodius University of Skopje
	Ruhr University Bochum
 National Observatory of Athens, Institute of Geodynamics, Athens	2Y Istituto Nazionale di Geofisica e Vulcanologia
	Istituto Nazionale di Oceanografia e di Geofisica Sperimentale
 National Institute of Oceanography and Applied Geophysics (OGS)	4P Istituto Nazionale di Geofisica e Vulcanologia
	Department of Applied Earth Sciences, Faculty of Geo-Information Science and Earth Observation, University of Twente
 Istituto Nazionale di Geofisica e Vulcanologia (INGV)	LabSis, Geo3Bcn-CSIC, Barcelona
 Croatian Seismological Survey (CSS)	7B Orfeus Data Center
 ORFEUS (KNMI) Data Center, Royal Netherlands Meteorological Institute	Croatian Seismological Survey
 UniWien (Department of Meteorology and Geophysics)	Department of Meteorology and Geophysics, University of Vienna
Norwegian Pool of Advanced Broad-Band Seismic Instruments	9H University of Bergen
 University of Bergen (UiB Norway)	Geophysical Department, University of Zagreb
 University of Zagreb	University of Oslo ... or as it is now:
	Geological Survey of Norway ... or as it is now:
	Norwegian Seismic Array ... or as it is now:
 ISTERre (ISTERRE)	
 Research Institute in Astrophysics and Planetology (IRAP France)	XP RESIF
	ISTerre (ISTERRE)
	Research Institute in Astrophysics and Planetology (IRAP France)
 Eidgenössische Technische Hochschule (ETH Zurich) - Swiss Seismological Service	Y5 Swiss Seismological Service at Eidgenössische Technische Hochschule Zürich
	Republic Hydrometeorological Service, Republika Srpska, Bosnia and Herzegovina
	Hydrometeorological Institute of Federation of Bosnia and Herzegovina
 National Institute for Earth Physics (NIEP Romania)	
	Y8 National Institute for Earth Physics, Romania
 Ludwig Maximilians Universität München (LMU-Germany)	National Institute of Geophysics, Geodesy and Geography, Bulgarian Academy of Sciences
	Sofia University "St. Kliment Ohridski"
	Institute of Geology and Seismology, Moldavian Academy of Sciences
	Department of Seismic Activity of Carpathian area (Lviv), Institute of Geophysics by S.I. Subbotin, National Academy of Sciences of Ukraine
	Institute of Rock Structure and Mechanics of the Czech Academy of Sciences
	Institute of Earth Sciences, University of Silesia
	Department of Earth Sciences, Uppsala University
	Sodankylä Geophysical Observatory, University of Oulu
	Institute of Seismology, Department of Geosciences and Geography, University of Helsinki
	Institute of Geophysics, University of Warsaw
	Institute of Geophysics, Czech Academy of Sciences
	Aarhus University
	Technische Universität Bergakademie Freiberg
	Institute of Geophysics, Polish Academy of Sciences
	Z6 Department of Earth and Environmental Sciences, Ludwig-Maximilian University Munich
	Earth Science Institute of the Slovak Academy of Sciences
	Department of Seismology, Institute of Geosciences, Polytechnic University of Tirana
	Seismological Network of Kosovo, Geological Survey of Kosovo
	Sector of Seismology, Institute of Hydrometeorology and Seismology of Montenegro
	Institute of Earth Physics and Space Science, Sopron and Budapest
	Institute of Geophysics, Czech Academy of Sciences
	Institute for Geosciences, University of Kiel
	Karlsruhe Institute of Technology

AdriaArray

AdriaArray - Logo

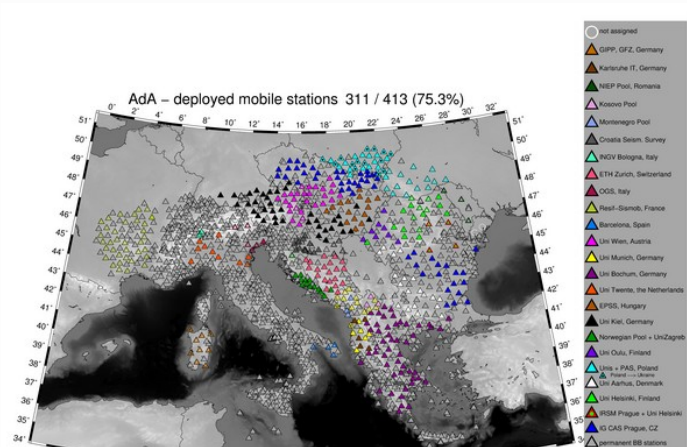
The logo will be discussed at the third AdriaArray international [workshop](#) in Dubrovnik, Croatia, on 3-5 April 2023.

AdriaArray - Mission

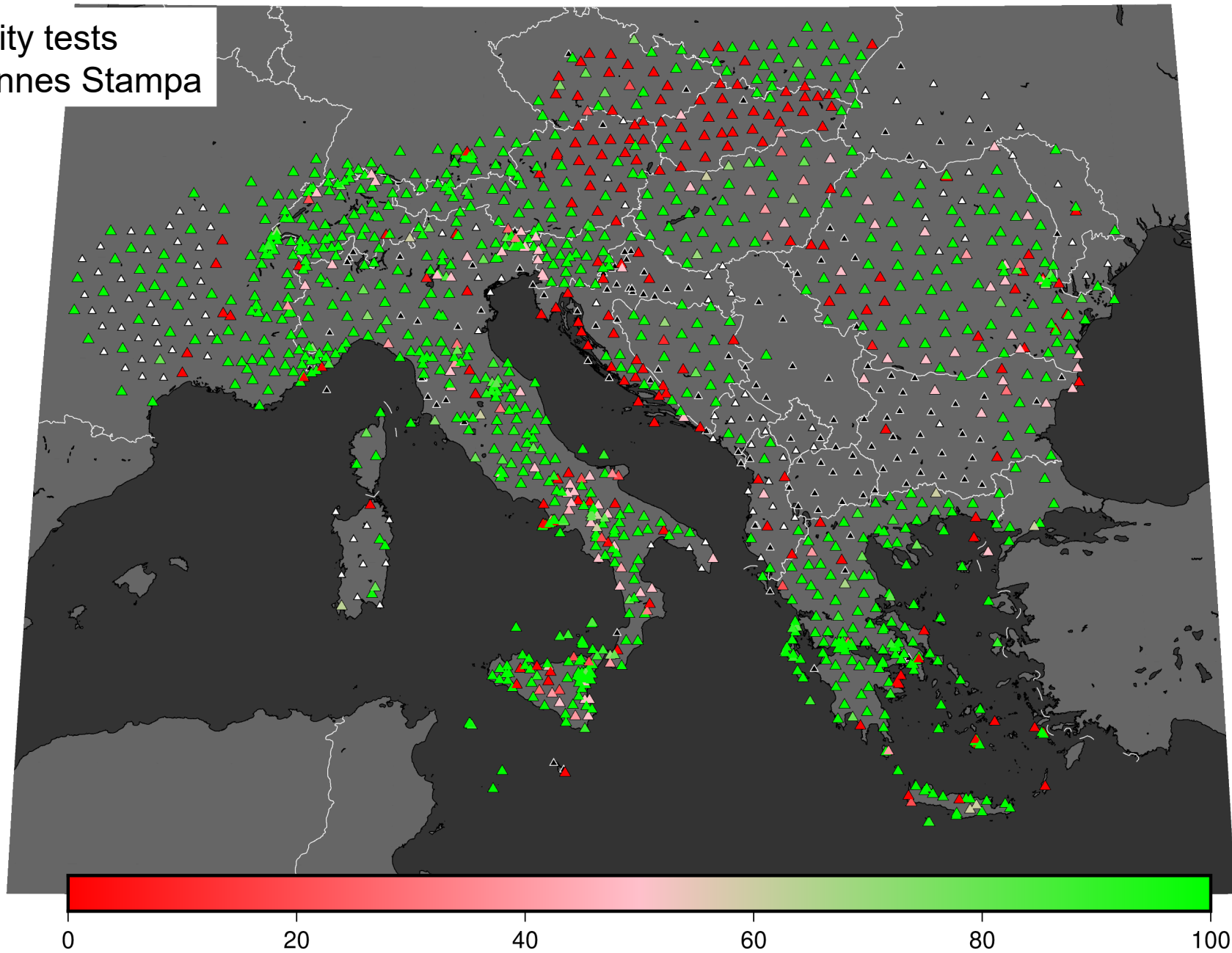
AdriaArray is a multi-national effort to cover the Adriatic Plate and its active margins in the central Mediterranean by a dense regional array of seismic stations to understand the causes of active tectonics and volcanic fields in the region. Plate-scale observations are complemented by local and LargeN experiments in key areas. The AdriaArray region reaches from the Massive Central in the west to the Carpathians in the east, from the Alps in the north to the Calabrian Arc and mainland Greece in the south. The deployment of seismic stations and scientific research is coordinated by the AdriaArray Seismology Group based on [FAIR](#) and open data exchange. Analyses of seismicity and multi-scale passive seismic imaging will lay the ground for a physical understanding and modelling of plate deformation and associated geohazards.

AdriaArray - Current deployment

Current status of the AdA seismic network. Filled triangelns represent stations already in the field, while empty triangles indicate stations that are not yet deployed.



availability tests
by Johannes Stampa



0

20

40

60

80

100

availability [%]

- △ not available during test
- △ not deployed yet

“online/offline” and “EIDA / nonEIDA”

1

station is **offline**, and **not in EIDA** ... bad case

2

station is **offline**, and **in EIDA**, data backfilled in batches

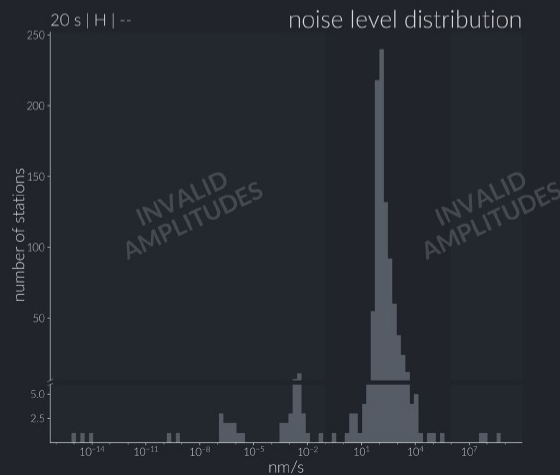
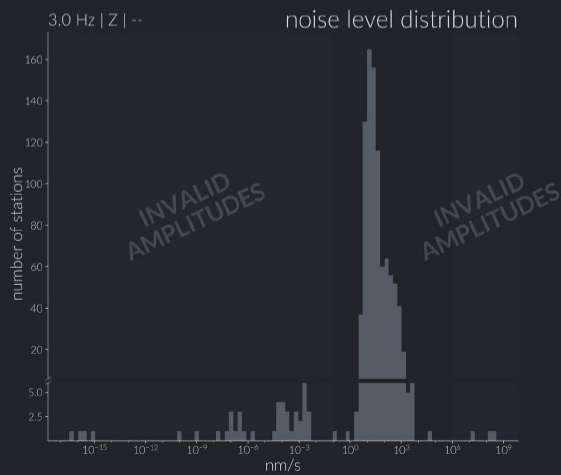
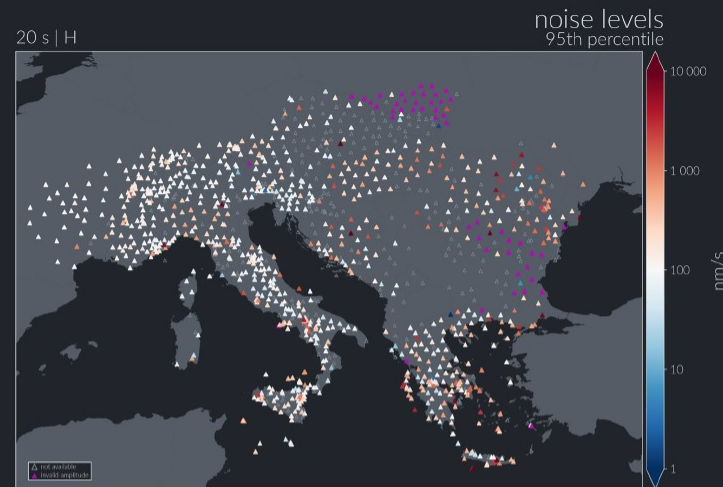
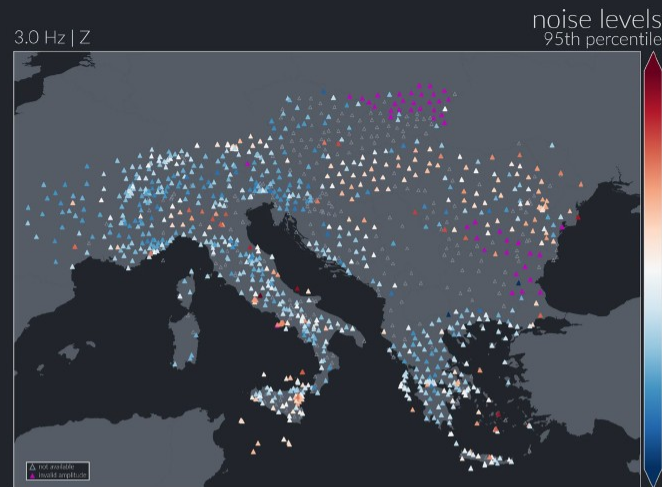
3

station is **online** and **not in EIDA** ... please connect it

4

station is **online** and **in EIDA** ... nice, thank you

noise maps by Felix Eckel

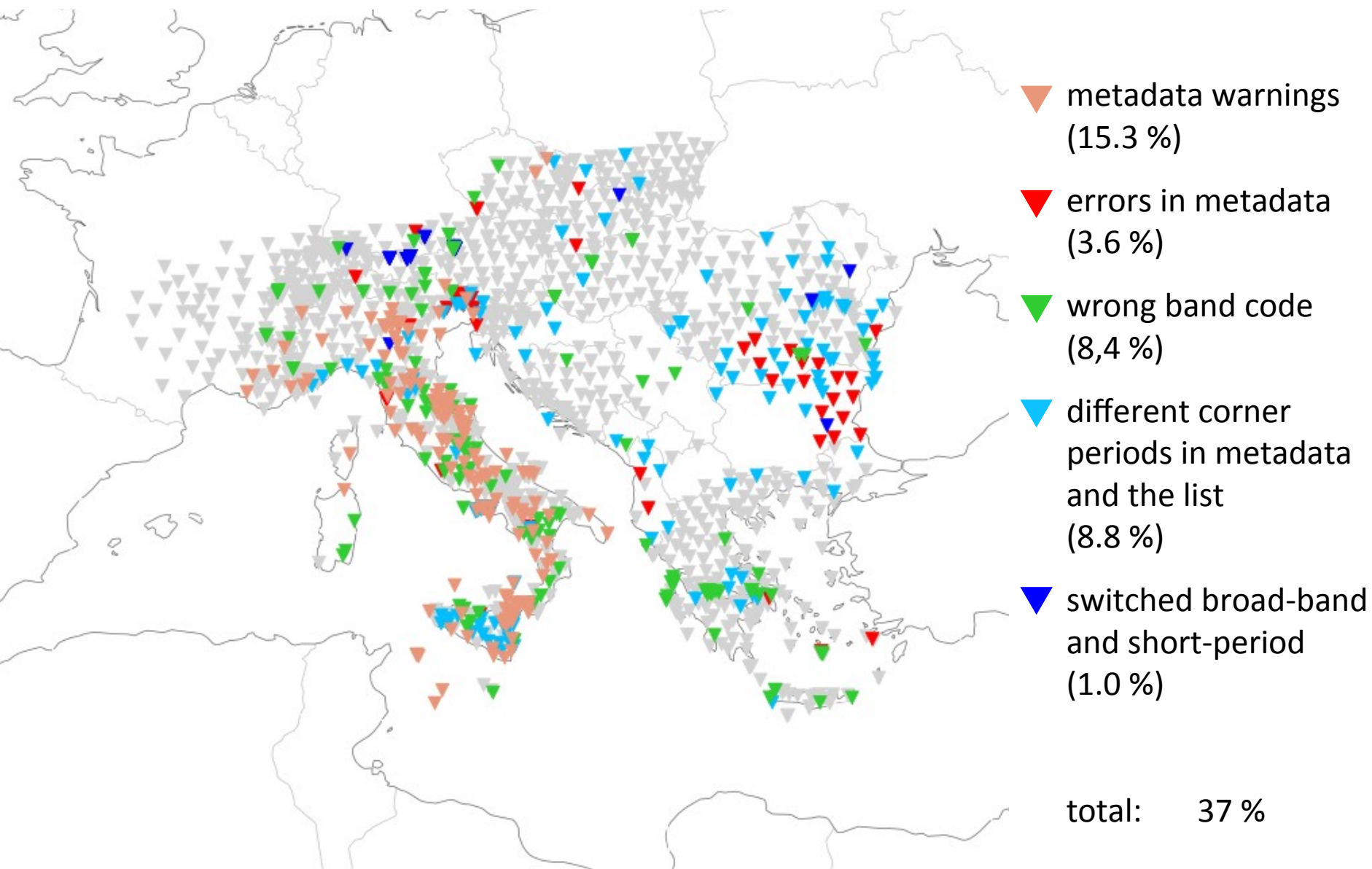


noise maps

- up to 72h of data
- detrend
- filtering
- 95th percentile of abs. amplitude
- color coding
- bad amplitudes:
< $1e-1$ or
> $1e6$

Metadata State-of-Health

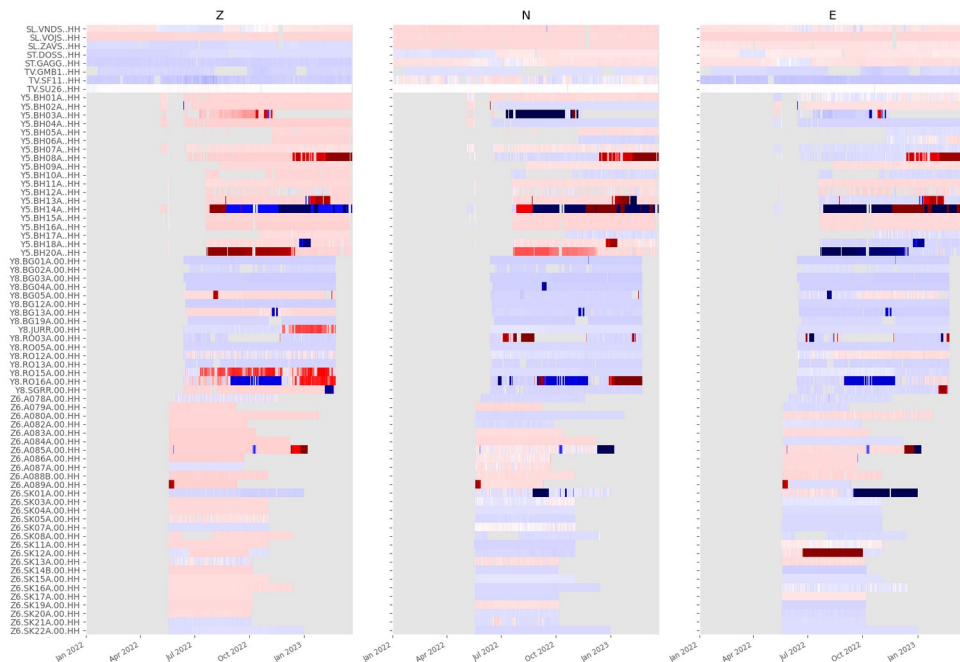
Testing: StationXML scheme, response, corner periods, band codes, P&Z consistency



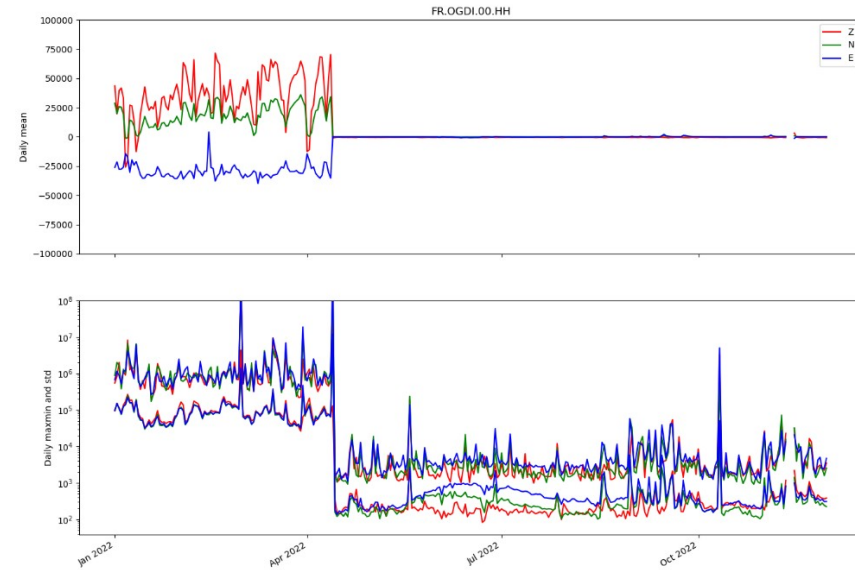
Daily means (medians)

- fast and powerful tool for QC
- metrics from EIDA WFCatalog:
daily median, daily std, daily max, min
- detection of saturated offsets, signal clipings

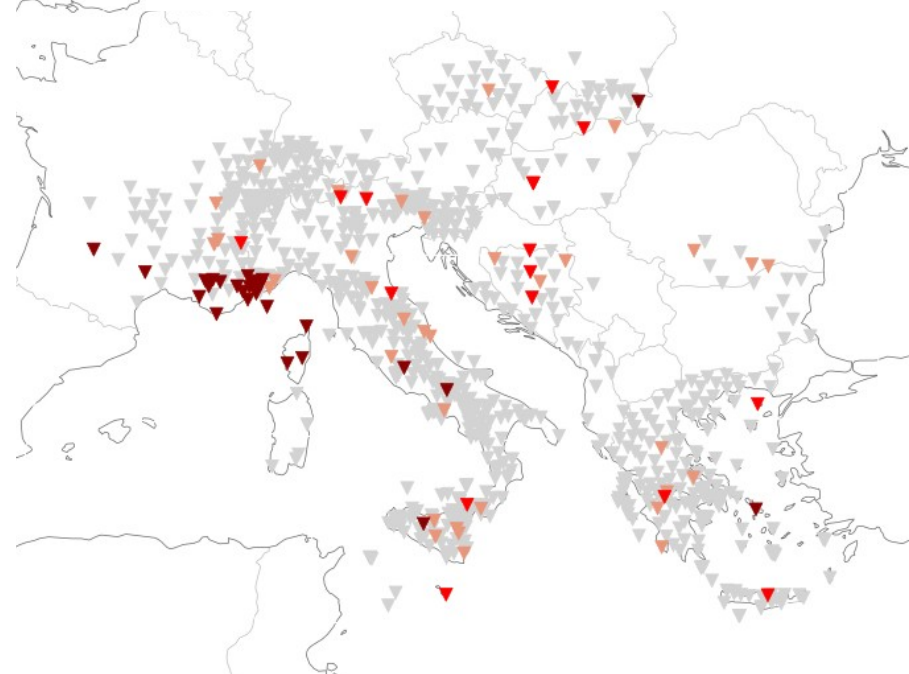
Detection of saturated offsets



Station State-of-Health

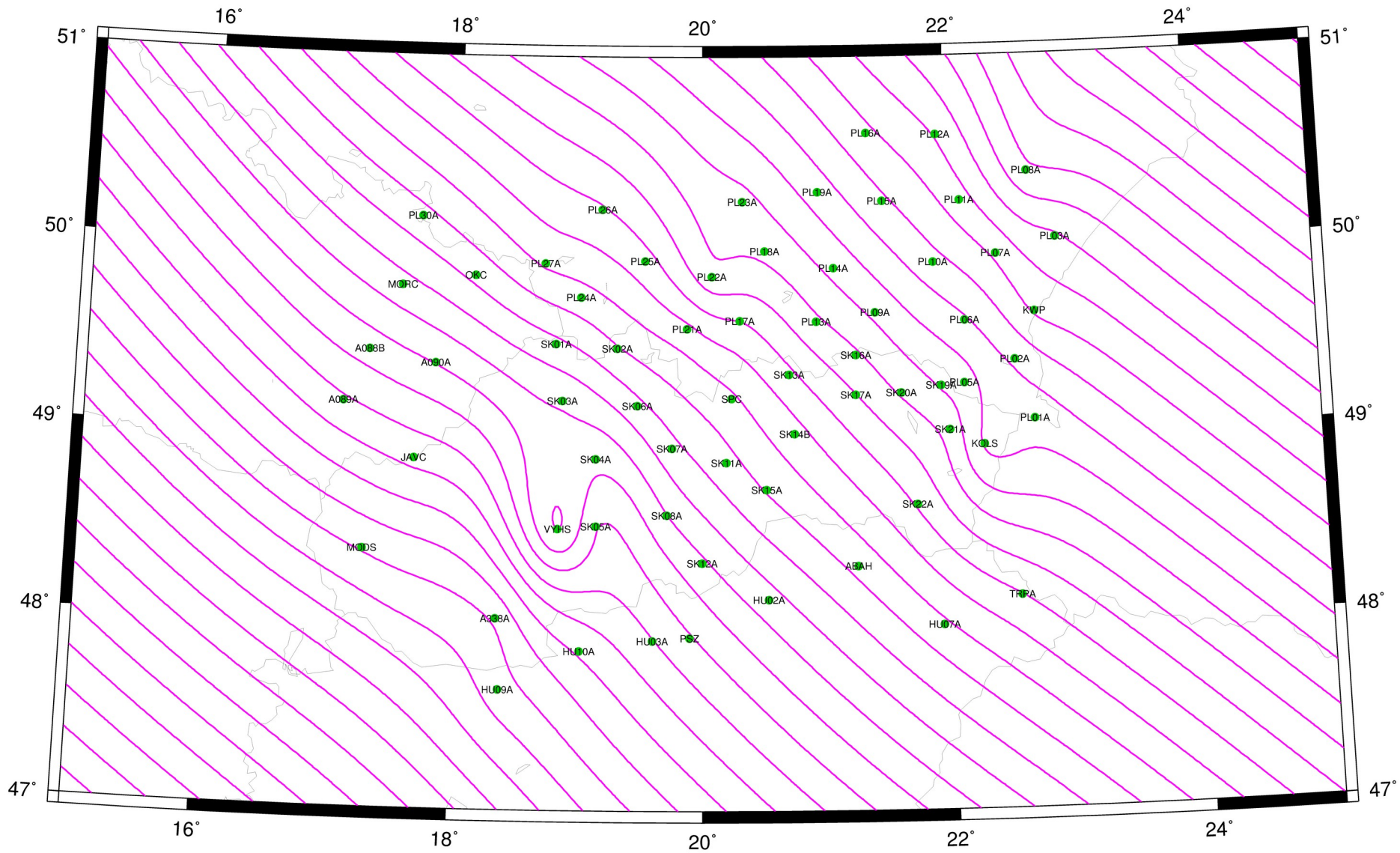


Frequency of saturated offsets



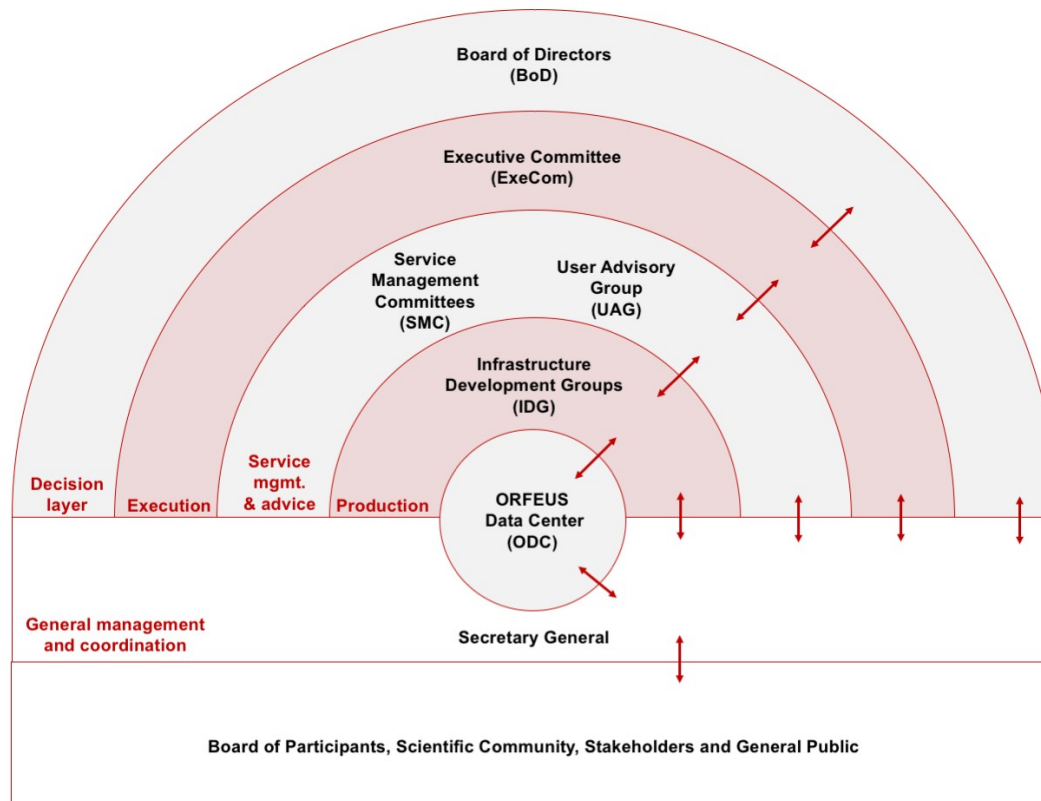
56 s

wavefront check by Petr



Orfeus User Advisory Group

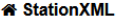
Petr Kolínský	(chair, Czech Republic, IG CAS)
Stéphane Rondenay	(Norway, Uni Bergen)
Dino Bindi	(Germany, GFZ)
Luděk Vecsey	(Czech Republic, IG CAS)
Antje Schlömer	(Germany, LMU)
Valerio Poggi	(Italy, OGS)
Davor Stanko	(Croatia, Uni Zagreb)
Felix Eckel	(Germany, Uni Kiel)



StationXML documentation

```
<FDSNStationXML schemaVersion="1.2">
  <Source>SeisComP</Source>
  <Sender>GFZ</Sender>
  <Created>2023-04-02T17:49:15.915266Z</Created>
  <Network code="Z3" startDate="2015-01-01T00:00:00Z" endDate="2022-12-31T00:00:00Z" restrictedStatus="open">
    <Description>
      AlpArray Seismic Network (AASN) temporary component
    </Description>
    <Identifier type="DOI">10.12686/alparray/z3_2015</Identifier>
  </Network>
  <Station code="A412A" startDate="2017-06-15T16:45:38Z" endDate="2018-02-12T09:54:14Z" restrictedStatus="open">
    <Comment id="1">
```

revision of 1.1 ==> 1.2 by
Philip Crotwell
Robert Casey
Jonathan Schaeffer
Petr Kolínský
accepted in 2022 by FDSN



v1.2

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 - 2.2.1. <Description>
 - 2.2.2. <Identifier>
 - 2.2.3. <Comment>
 - 2.2.4. <DataAvailability>
 - 2.2.5. <Operator>
 - 2.2.6. <TotalNumberStations>
 - 2.2.7. <SelectedNumberStations>
 - 2.3. <Station>
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 - 2.5. <Response>
- 3. Specifying and Using Response Information
- 4. StationXML Tools
- 5. Appendices
 - Specification home
 - FDSN home

Read the Docs

v: v1.2

2.2. <Network> required

Warning, Future Change

endDate: This attribute should not be used if it is in the future.

Warning, Future Change

endDate: This attribute is likely to require timezone of Z.

Warning, Future Change

startDate: This attribute is likely to require timezone of Z.

The Network container. All station metadata for this network is contained within this element. A Description element may be included with the official network name and other descriptive information. An Identifier element may be included to designate a persistent identifier (e.g. DOI) to use for citation. A Comment element may be included for additional comments.

An active Network should not use the endDate attribute. Unlike SEED, do not use an endDate in the distant future to mean active.

Example: <Network code="XX" startDate="2016-01-27T13:00:00Z" />

Attributes of <Network>:

attribute	type	required	description	example
alternateCode	string	no	A code used for display or association	alternateCode="GSN"
code	string	yes	Name of Network	code="XX"
endDate	dateTime	no	End date of network. Do not use if still active, endDate should not be in the future.	endDate="2018-01-27T00:00:00Z"
historicalCode	string	no	A previously used code if different from the current code	historicalCode="XX"
restrictedStatus	RestrictedStatusType	no	One of: "open", "closed", "partial"	restrictedStatus="open"
sourceID	anyURI	no	A data source identifier in URI form. It is recommended that this follow the FDSN Source Identifiers specification, http://docs.fdsn.org/projects/source-identifiers	sourceID="FDSN:XX"
startDate	dateTime	no	Start date of network	startDate="2016-07-01T00:00:00Z"

Sub Elements of <Network>:

element	type	number
<Description>	string	optional

suggestion by Costas Papazachos

- we can include in the description of the installed stations information about geology, etc.

Station Code	Geology					Geophysical Structure				
	Age	Description	Geologic Age [M: Mesozoic and older, T: Tertiary, Q: Quaternary (undifferentiated), P: Pleistocene, H: Holocene]	Bedrock (Yes, if formation is compact, e.g. Mesozoic, Tertiary limestones and basalts, etc., typically $V_{so} > 600\text{--}800\text{m/s}$)	Geological Map Scale	V_{so} (m/s)	VS30 (m/s) Determination A: Direct investigation (Cross-Hole, Down-Hole, SW, Refraction, Ambient noise, Other) - Define method B: From investigations in the same area and same material C: From investigations in the same formation D: From literature (provide reference)	Bedrock depth (m)	V_z (m/s): Average velocity determined from a Vs profile up to the maximum depth of investigation, Z(m)	Maximum depth of investigation, Z(m)
LSM0	Mesozoic	Leucocratic albite-sericite-microcline gneiss	M	Yes	1:50,000					
PIR1	Holocene	Alluvial deposits	H		1:50,000	420	A (Down-hole)	50	590	55
STL1	Upper Miocene - Lower Pleistocene	Sandstone-marl series sandstones friable to quite compact, locally microconglomerates with cross-bending are interbedded. At places there are marl horizons.	P		1:50,000					
AIA1	Pleistocene-Pliocene	Clastic Formations	P		1:50,000					
KYP2	Quaternary-Holocene	Alluvial deposits	H		1:50,000	652	A (Cross-hole)			
DRA2	Pleistocene	Cohensive conglomerates	P		1:50,000					
ART2	Upper Senonian	Brecciated limestone	T	Yes	1:50,000					
NAX1	Mesozoic / Early Tertiary	Granodiorite of quartz, feldspar, plagioclase, biotite, and hornblende; clinopyroxene may also occur	M	Yes	1:50,000					
SIA1	Miocene (Upper Aquitanian-Tortonian)	Alternating conglomerates, sandstones and marls	T		1:50,000					
ITS1	Upper Miocene - Lower Pleistocene	Red clay series red to brick red, silty clays, with mica and calcareous concretionary bodies	P		1:50,000					
KSS1	Quaternary	Alluvial deposits	Q		1:50,000					
AIG2	QUATERNARY	Old and new talus cones. Scree	Q		1:50,000					
PLN1	Quaternary	Alluvial deposits	Q		1:50,000					
KMT1	Quaternary	Alluvial deposits	Q		1:50,000	253	A (MASW)			
KLR1	Upper Miocene - Lower Pleistocene	Red clay series red to brick red, silty clays, with mica and calcareous concretionary bodies	P		1:50,000					
LMS2	Holocene	Old terrestrial terraces	H		1:50,000					
KOZ2	Lower to Upper Pliocene	Deposits of marls, claystones, sands, and coal covered by adjacent quaternary formations; progressively layers with lenses of poorly cemented conglomerates	T		1:50,000	421	D (Meier et al., 2018, SRL)			
LMN1	Holocene	Alluvial deposits	H							
AST1	Quaternary-Holocene	Recent deposits: Sand gravel and rounded pebbles	H		1:50,000					

TOKEN to get the data

EIDA Authentication System = EAS

<https://geofon.gfz-potsdam.de/eas/EIDAAuthenticationService.pdf>

manual, 8 pages, easy to read

EAS User Documentation

Release 0.9b1

Javier Quinteros and Andres Heinloo

May 16, 2019

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<https://geofon.gfz-potsdam.de/eas/>
Request token

Epos



User documentation with all the details can be found in the following [link](#).

EIDA users requesting Alarray data must complete all the requirements mentioned below for the registration process and later get in contact with the Network PI to be authorized to access the data.

From this page you can request a digitally signed token to be used with all existing EIDA web services (not Arclink) in order to not only retrieve open or restricted data, but also personalize your interaction with the EIDA services.

Please, select a duration for your token. After this amount of time, the token will not be accepted anymore. After clicking on the "Request token" button you will be redirected to B2ACCESS (optionally your home institution) to complete the authentication.

1 day
2 days
1 week
2 weeks
1 month

Request token

create an account at B2ACCESS, if not existing: "No account? Sign up"
<https://b2access.eudat.eu/saml-idp/saml2idp-web-entry>

"Select group:" and [request to be part of EPOS](#) !!!

GO TO EUDAT WEBSITE



[WHAT IS B2ACCESS](#) [USER GUIDE](#) [FAQS](#) [CONTACT](#)

[No account? Sign up.](#)

[Forgotten password?](#)

-  Sign in with Facebook
-  Sign in with GitHub
-  Sign in with Google
-  Sign in with INDIGO IAM
-  Sign in with Microsoft Live
-  Sign in with ORCID

-
-  Institute of Engineering Seismology and Earthquake Engineering
 -  UNIVERSIDAD TECNOLÓGICA DE PEREIRA
 -  University of Jinan
 -  29 Mayis University
 -  A*STAR - Agency for Science, Technology and Research
 -  A. T. Still University
 -  AAF Virtual Home
 -  aai.lab.maeen.sa
 -  AAI@EduHr Single Sign-On Service
 -  Aalborg University

[Cancel authentication](#)

- Send an email to Cédric Legendre and request to be added to the AdriaArray group. Let him know which email you registered.
- Upon confirmation, you should be able to request a token that gives access to the AdriaArray data.

Epos



User documentation with all the details can be found in the following [link](#).

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1 day
2 days
1 week
2 weeks
1 month

Request token

GO TO EUDAT WEBSITE

WHAT IS B2ACCESS USER GUIDE FAQs CONTACT

[No account? Sign up.](#)

Search

User name
Password
Sign in
[Forgotten password?](#)

Sign in with Facebook
Sign in with GitHub
Sign in with Google
Sign in with INDIGO IAM
Sign in with Microsoft Live
Sign in with ORCID

Institute of Engineering Seismology and Earthquake Engineering
UNIVERSIDAD TECNOLÓGICA DE PEREIRA
University of Jinan
29 Mayis University
A*STAR - Agency for Science, Technology and Research
A. T. Still University
AAF Virtual Home
aalab.maaen.sa
AAI@EduHr Single Sign-On Service
Aalborg University

[Cancel authentication](#)

token example:

A screenshot of a Notepad window titled "Lister - [G:\AlpPhase\eidatoken]". The window has a menu bar with "Soubor", "Editace", "Možnosti", and "Nápověda". The text content is a PGP signed message. It starts with "BEGIN PGP MESSAGE" and "-----BEGIN PGP SIGNED MESSAGE-----". The hash is "SHA1". The JSON payload contains fields: "valid_until" (2023-03-13T00:42:50.392840Z), "cn" (Petr Kol\u00edednsk\u00fd), "memberof" (/epos/alparray;/epos;/), "sn" (Kol\u00edednsk\u00fd), "issued" (2023-02-11T00:42:50.392844Z), "mail" (petr.kolinsky@univie.ac.at), and "givenName" (Petr). The expiration is "1m". It then shows "-----BEGIN PGP SIGNATURE-----" and "Version: GnuPG v1". The signature block is a long base64-encoded string. It ends with "-----END PGP SIGNATURE-----".

```
Lister - [G:\AlpPhase\eidatoken]
Soubor Editace Možnosti Nápověda 100 %
BEGIN PGP MESSAGE
-----BEGIN PGP SIGNED MESSAGE-----
Hash: SHA1

{"valid_until": "2023-03-13T00:42:50.392840Z", "cn": "Petr Kol\u00edednsk\u00fd",
"memberof": "/epos/alparray;/epos;/", "sn": "Kol\u00edednsk\u00fd", "issued":
"2023-02-11T00:42:50.392844Z", "mail": "petr.kolinsky@univie.ac.at", "givenName":
"Petr", "expiration": "1m"}
-----BEGIN PGP SIGNATURE-----
Version: GnuPG v1

iQEcBAEBAGAGBQJj5uSKAAoJEEFpzp0A1wdXKPwH/1DryAijRfMTQh2uP+/JNH1Y
Cd2SbQyucQW+SA00FupUYNkzTbiyUkiNnd9cShBIZ0ea3rDpP4Ht6RQah0SHjEYY
EhY/1/w6Jmpf85KpkSi3p02r3LNhbUjksRfnI9giDUzp5lMcgaAgp+ftKwuL1IJ
0e+3cDoxyXPuKo9XP19RURr/DG3l+iQ+3xQYR1ZUS5qpRi25T16c3L8vQPXnH+n8
E8/gx3/otatUTidTn5H2aCXAZolorM005A1ArC25TuKxze7ZDEo+S/NIZr354KDJ
gnmNgu0XB1LIYUgw/sZGPC1up3pqCU2iaURFu1NQogSDXTnpDYkFI/ICLeA2MAc=
=Hc/f
-----END PGP SIGNATURE-----
```

data / metadata download

web browser metadata:

🔒 <https://eida-sc3.infp.ro/fdsnws/station/1/query?net=Y8&station=RO03A&channel=???&level=response>

Tento XML soubor nemá připojeny žádné informace o vzhledu prvků. Strom XML dokumentu je zobrazen níže

```
—<FDSNStationXML schemaVersion="1.1">  
  <Source>SeisComP</Source>  
  <Sender>NIEP</Sender>  
  <Created>2023-04-02T18:09:07.692264</Created>  
  —<Network code="Y8" startDate="1980-01-01T00:00:00" restrictedStatus="closed">  
    <Description>Adria Array Network</Description>  
    —<Station code="RO03A" startDate="2022-06-06T09:00:00">  
      <Latitude>44.77653</Latitude>
```

web browser data (open):

🔍 <https://geofon.gfz-potsdam.de/fdsnws/datasetselect/1/query?network=CZ&station=KHC&start=2023-01-01T00:00:00&end=2023-01-01T23:59:59>

web browser data (embargo):

ObsPy:

RoutingClient("eida-routing", credentials={'EIDA_TOKEN': './eidatoken.pgp'})

using the virtual network _ADARRAY (example by Luděk)

```
In [23]: rclient_eida = RoutingClient("eida-routing", credentials={'EIDA_TOKEN': token})
inv_eida_ada = rclient_eida.get_stations(network="_ADARRAY", level="station", includerestricted=True)
print(inv_eida_ada)
```

Inventory created at 2023-03-29T21:48:04.182604Z

Created by: ObsPy 1.4.0

<https://www.obspy.org>

Sending institution: ObsPy FDSN Routing 1.4.0, RESIF-SI, SeisComp (BGR, GFZ, KOERI, LMU, NIEP, NOA, ODC, RESIF-DC, SED, UIB)

Contains:

Networks (35):

1Y, 7B, 9H, BS, BW, C4, CH, CL, CR, CZ (2x), FR, G, GE, GR, HA, HC
HL, HP, HT, HU, KO, MD, ME, MT, OE, PL, RD, RO, SK, SL, XP, Y5,
Y8, Z6

Stations (893):

1Y.GR01 (Neo Petritsi, Serres, Greece)
1Y.GR02 (Drama, Greece)
1Y.GR03 (Xanthi, Greece)
1Y.GR04 (Abdera, Greece)
1Y.GR05 (Maronia-Sapes, Petrota, Greece)
1Y.GR06 (Mavrokklisi, Greece)
1Y.GR07 (Mesimeri, Edessa, Greece)
1Y.GR09 (Rentina, Greece)
1Y.GR10 (Kriovrisi, Ptolemaida, Greece)

```
In [ ]: data_temp = rsClient.get_waveforms(network=net.code, station=sta.code, location='*', channel=cha.code,
starttime=start, endtime=end)
```

ObsPy:

```
RoutingClient("eida-routing", credentials={'EIDA_TOKEN': './eidatoken.pgp'})
```

fdsnws_fetch (token is assumed to be ~/.eidatoken

```
$ fdsnws_fetch -vvv -N Z3 -C "HHZ" -s "2016-03-01" -e "2016-03-01T00:02:00" -o data.mseed
```

webinterface:

<http://eida.gfz-potsdam.de/webdc3/>

and EIDA Data Archives

GEOFON | EIDA | Imprint | Data Protection

EIDA

GFZ

Helmholtz Centre
POTSDAM

Explore events

Explore stations

Submit request

Download data

View console

[doc Help](#)

Stations Controls

Station Information

Browse Inventory

User Supplied

Networks

Year from 1980 to 2023:

Network Type:

All nets

Network Code:

All Networks

* = temporary network; += restricted access

Stations

by Code

by Region

by Events

Filter stations by station code:

All Stations

Streams

by Code

by Sampling

Choose the desired set of channels:

Use SHIFT and CTRL to extend the set

Event and Station Map

Use left SHIFT + drag mouse to select regions.

[Legend Help](#)

Event and Station List

<https://orfeus-eu.org/webdc3/>

orfeus

Home

Organization

Data & Services

Other

ORFEUS Data Center WebDC3 Web Interface

Explore events

Explore stations

Submit request

Download data

Console

[Information](#)

Events Controls

Use this to select events to compose your request.

Event and Station Map

Here the events and stations you choose will be displayed. Use the mouse to drag the map around. Use the Ctrl-Mouse to draw areas for limiting the search of events and stations when the appropriate modules are enabled.

Event Information

Catalog Services

User Supplied

Catalog Service: GFZ

Date Interval (yyyy-mm-dd): 2023-03-26 - 2023-04-02

Minimum Magnitude: 3

Depth from 0 to 999 km

Coordinates: (Use -ve for SW, +ve for NE)

N

90

W -180 180 E

-90

S

Clear

Event and Station List

Use these tables to check which events and stations have been selected by your search criteria. You can interactively remove unwanted items. Your final request will be built from the currently displayed information.

Data availability & access, quality tests, stationXML, AdA webpage

Q (& A)

@ \tomorrows\breakout\session

AdriaArray Workshop, April 3-5, 2023, Dubrovnik

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INSTITUTE OF GEOPHYSICS
OF THE CZECH ACADEMY OF SCIENCES

